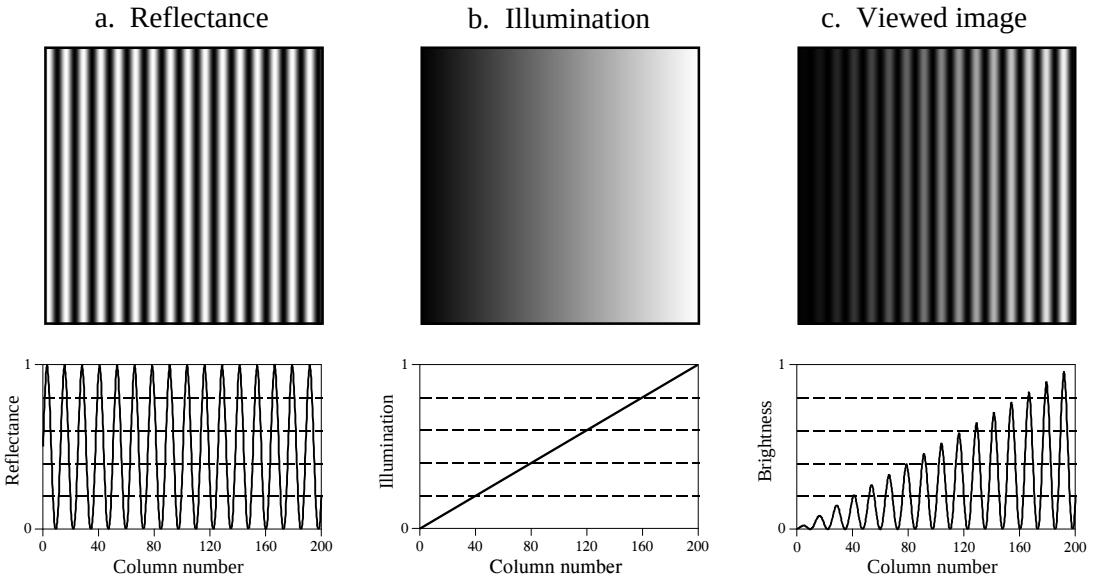


FIGURE 24-8
Model of image formation. A viewed image, (c), results from the multiplication of an illumination pattern, (b), by a reflectance pattern, (a). The goal of the image processing is to modify (c) to make it look more like (a). This is performed in Figs. (d), (e) and (f) on the opposite page.

in an approximation to the original illumination signal, (b). This is where convolution by separability is used. The exact shape of the PSF is not important, only that it is much wider than the features in the reflectance image. Figure (d) is the result, using a Gaussian filter kernel.

Since a smoothing filter provides an estimate of the illumination image, we will use an edge enhancement filter to find the reflectance image. That is, image (c) will be convolved with a filter kernel consisting of a delta function minus a Gaussian. To reduce execution time, this is done by subtracting the smoothed image in (d) from the original image in (c). Figure (e) shows the result. It doesn't work! While the dark areas have been properly lightened, the contrast in these areas is still terrible.

Linear filtering performs poorly in this application because the reflectance and illumination signals were original combined by multiplication, not addition. Linear filtering cannot correctly separate signals combined by a nonlinear operation. To separate these signals, they must be *unmultiplied*. In other words, the original image should be *divided* by the smoothed image, as is shown in (f). This corrects the brightness and restores the contrast to the proper level.

This procedure of dividing the images is closely related to **homomorphic processing**, previously described in Chapter 22. Homomorphic processing is a way of handling signals combined through a nonlinear operation. The strategy is to change the nonlinear problem into a linear one, through an appropriate mathematical operation. When two signals are combined by